GGS 662 AGENTIC GEOAI · ASSIGNMENT 4 · PROGRESS

Seeing Through Clouds

What a flood map can reliably say about roads it cannot see

HURRICANE IDA 2021 · SPACENET 8 · 0.5 M · GREYSCALE

Justin Rogers · George Mason University · 30 October 2026

Preliminary results on development data · Imagery: Maxar via SpaceNet 8



The question, refined since Assignment 3

THE PROBLEM

"Clouds can hide target areas in images taken after disasters. This can lead to labels being inaccurate compared to ground truth due to obscured parts of the imagery. How should we detect clouds in this imagery, handle them, and report them? The goal is results, such as maps of flooded roads, that people can trust."

Since Assignment 3. Assignment 3 asked what an agent adds to reading change. Re-running that agent workflow showed the first thing a system must decide: can it see the ground at all?

Expected contribution: a tested rule for what a change map may say under cloud, a measured label gap, and a test of whether a second date can correct thin cloud.

LEAD · REPORT · Q2

What can a flood-road map reliably tell users about roads hidden by cloud?

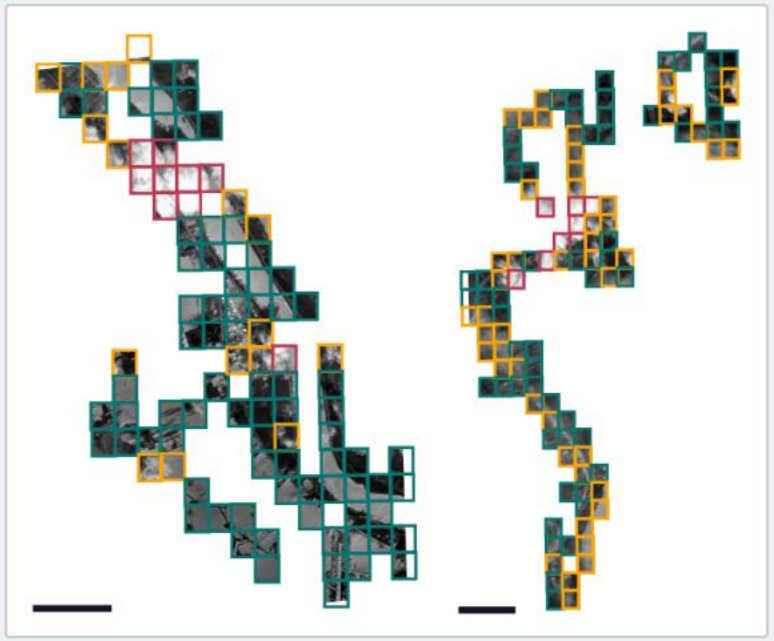
HANDLE · Q3

When only the before image is cloudy, does the after image alone find change better than both images together?

DETECT · Q1 · THE TEST BED

Can we use synthetic clouds to represent real-life blockages in satellite imagery?

Study area and data



clear thin cloud or haze opaque

South scene (102 tiles) and west scene (115 tiles), after images. Black bar = 2 km.

Imagery	Maxar before and after, 0.5 m, Hurricane Ida 2021; used in greyscale
Roads	1,037 OpenStreetMap roads with SpaceNet 8 flood labels (Hänsch et al. 2022), on 217 of the 226 development tiles
Split	186 train · 40 validation · 40 test tiles, plus a 333-tile north negative set; the test tiles, the negative set, Germany and Louisiana-West stay sealed
Cloud labels	HRC_WHU (Li et al. 2019): 150 Google Earth images, 0.5–15 m, expert cloud masks; 120 development, 30 sealed; academic use only
Quality issues	Flood labels are “yes” or missing, never “no” · 17 tiles under opaque cloud · one after image for all development tiles

Workflow: detect, handle, report

GGS 680 · CLASSICAL

1 · Detect

Greyscale masks on each date: bright cores (our rule), the dark channel (He et al. 2011), and its rise from before to after (our rule).

GGS 662 · THE AGENT GATE

2 · Handle, road by road

ASSESS	clear sky
CORRECT	thin cloud + clear second date
CANNOT ASSESS	opaque cloud

Correct = two-date de-hazing (our rule, after He et al. 2011 and Canty & Nielsen 2008), then the detector again. Filled pixels count as not observed.

WHAT THE MAP MAY SAY

3 · Report

Each statement about a hidden road gets a tier:

RELIABLE

SYNTHETIC ONLY

RANGE ONLY

CHANGE DETECTOR

Random forest (Breiman 2001) on classical, texture (Haralick 1979) and DINOv2 (Oquab et al. 2024) cues: 83.0% on 798 roads.

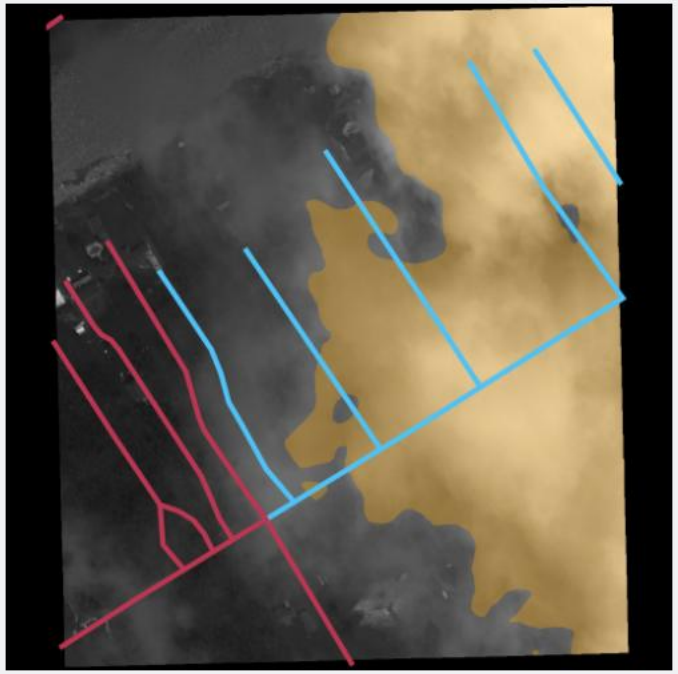
BASELINES AND CHECKS

Tile cloud rule; no correction. Tile-blocked folds (Roberts et al. 2017), McNemar tests (McNemar 1947), one sealed check.

AGENTS PROPOSE, I DECIDE

Role agents draft and critique. Decided by hand: the wording, the order Q2, Q3, Q1, the road cloud rule.

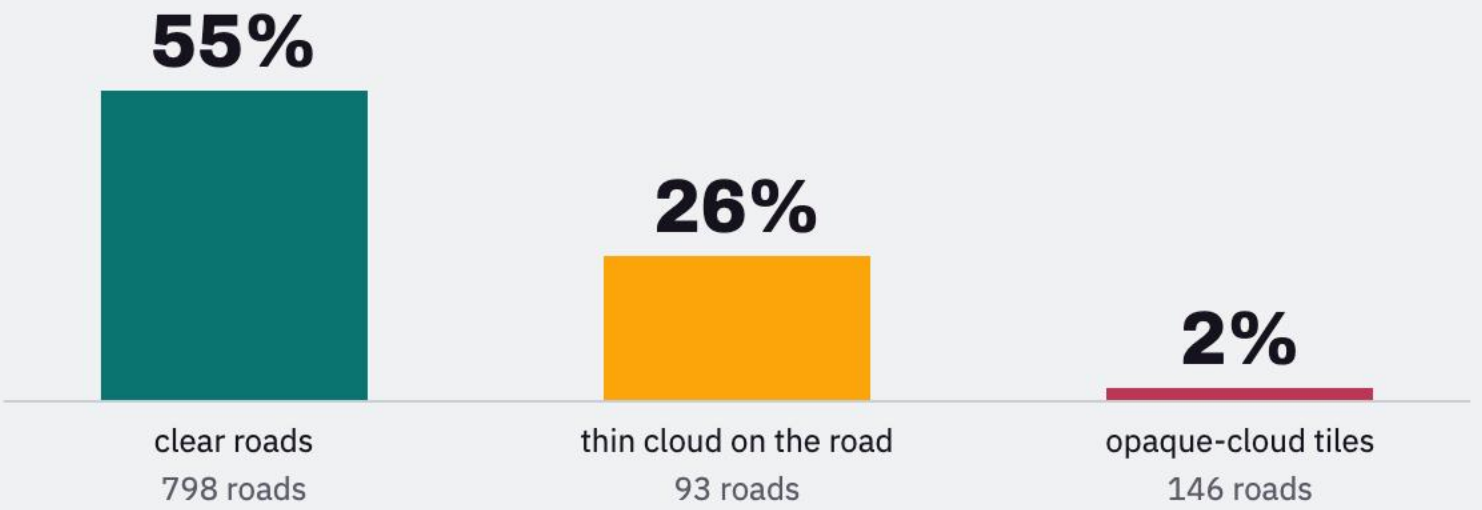
Under cloud, roads rarely get a flood label



labelled “flooded” thin cloud

no flood label (reads as dry)

ROADS LABELLED “FLOODED”, BY CLOUD ON THE ROAD



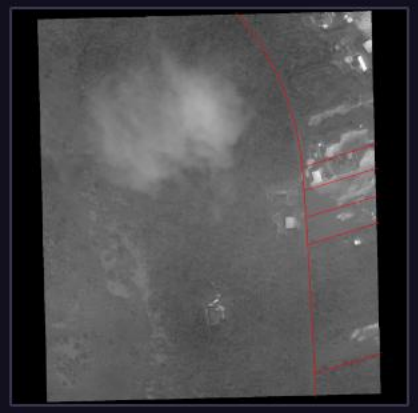
Analysts “only attribute what was visible” (Hänsch et al. 2022), so a flooded road under cloud reads as dry.

Part of the gap is geography: the south scene is dry around its cloudy tiles. Left: tile 2_15_45, after image · Imagery: Maxar via SpaceNet 8

The second image may help under thin cloud



BEFORE · CLEAR



AFTER · THIN CLOUD



AFTER · CORRECTED

+3 of 24

flooded roads under thin cloud found that the uncorrected image missed; none lost (71% → 83%; McNemar p = 0.25). 3 of 69 dry-labelled roads flipped to flooded.

Clear roads about the same: 77.7% → 78.3% on 175 (6 calls fixed, 5 broken). Single-image de-hazing (He et al. 2011): 88% found, clear roads −4 points.

Tile picked by a fixed rule: the corrected road under the most cloud. Two-date de-hazing: our rule, after He et al. (2011) and Canty & Nielsen (2008). Imagery: Maxar via SpaceNet 8 · red = labelled flooded.

WHAT GOES WRONG

At 0.5 m the two dates barely agree pixel by pixel (r = 0.46 here), so the fit reads clear ground as hazy and skips the bright core.

NEXT: FIX, THEN PRE-REGISTER

Match the dates on unchanged pixels at a coarser scale with a nonlinear tone match; then test once on unseen roads.

Feasibility, limits and next steps

Feasible

- Data, masks, de-hazer and the held-out harness all run
- Dark-channel mask: F1 0.65 against 0.41 for the old rule (HRC_WHU)
- Label gap measured: 55%, 26%, 2% labelled flooded (partly geography)

Limits

- One event, one sensor, roads only; all real cloud in one after image
- Sealed test tiles sit next to development tiles
- Only 24 trusted flooded roads under thin cloud
- De-hazing is exploratory and its fit is rough; the method critic says revise

Plan to the paper · 8 weeks

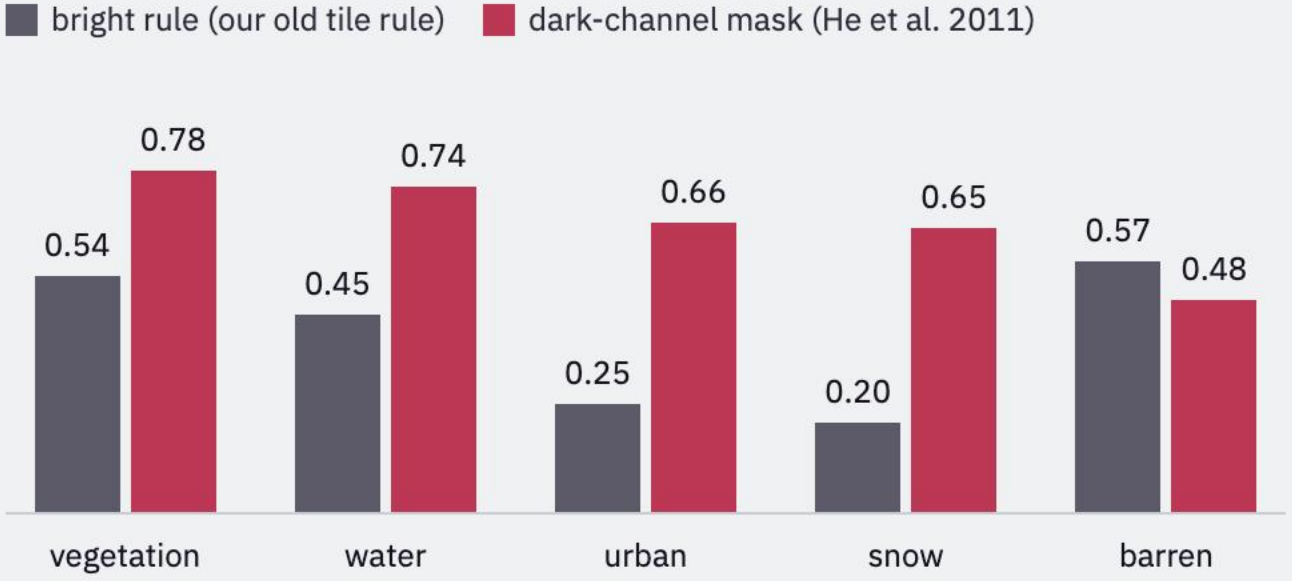
- Wk 1–2: pre-register; freeze the cloud generator and de-hazer
- Wk 3–4: realism checks, synthetic test bed (Q1)
- Wk 5: before-image cloud (Q3)
- Wk 6: report tiers and maps (Q2)
- Wk 7: one sealed look
- Wk 8: write and submit

QUESTION FOR THE CLASS

When a map cannot see a road, would you rather it say “cannot assess”, show a range, or fill it in from another date?

Cloud masks against human cloud labels

F1 AGAINST EXPERT CLOUD MASKS, BY LAND COVER



0.65

F1, dark-channel mask (precision 0.62, recall 0.68)

0.41

F1, bright rule (precision 0.81, recall 0.28)

120 HRC_WHU development images (Li et al. 2019; 30 sealed). Single-date masks only: the road rule's two-date mask needs a before image.

MASK-BIAS CHECK

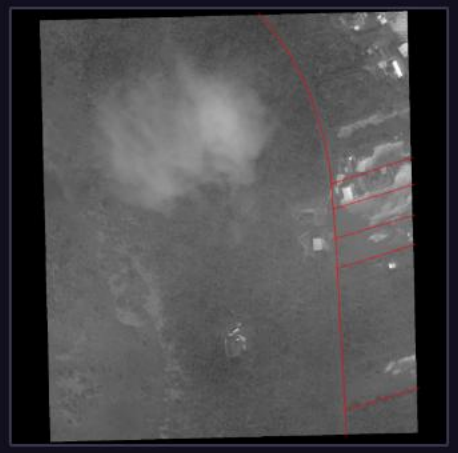
Synthetic haze on 769 clear roads: under thin haze the mask flags flooded and dry roads alike; under thick haze it flags flooded roads more (up to +13 points), not less.

How the two images are layered



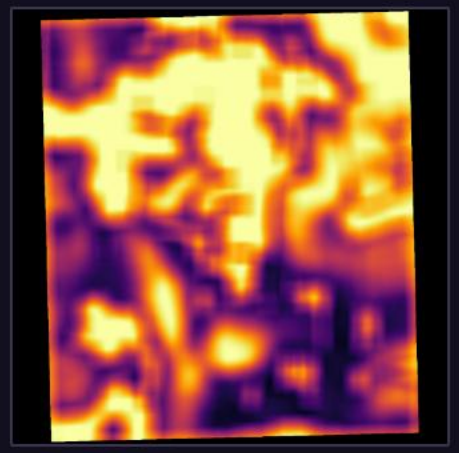
1 · BEFORE

Clear reference: the ground



2 · AFTER

Ground dimmed by t, plus light L



3 · FITTED t

Dark = treated as haze; bright = left alone



4 · CORRECTED

(after − L) ÷ t; red = labelled flooded

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after ≈ t·ground + L · ground ≈ g·before + h · corrected = (after − L) ÷ t
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Haze model as used by He et al. (2011). By design, changed pixels (floodwater) drop out of the fit (untested); t below 0.2 counts as opaque. Here r = 0.46, so the core is skipped and clear ground reads as hazy.

Critical human review and AI use

DECISION LOG

CHANGED	Problem statement: rewritten in Justin's own words
CHANGED	R1: DINOv2-vs-classical frame dropped; cloud generation kept
REJECTED	R3: the 20% cut-off as the question's frame
CHANGED	Q1 moved last, in Justin's wording
CHOSEN	Q2 leads (Claude had suggested Q3)
ACCEPTED	Road cloud rule for fitting and scoring
DEFAULT	Methodology critique: taken by Claude as defaults; Justin can overrule

AI USE

Claude Code (Anthropic) agents drafted research questions, critiques, literature notes, analysis code, figures and these slides.

Justin made the decisions on the left, except the one marked DEFAULT.

Every result on these slides was produced by code on development data; dataset facts come from the datasets' own documents. A validation agent checked the claims against the papers.

Agreement between agents is not treated as validation.

References

CITED ON THESE SLIDES · JIF ABOVE 4 OR COMPANY-LED

Li et al. 2019 HRC_WHU cloud masks · ISPRS Journal of Photogrammetry and Remote Sensing

He, Sun & Tang 2011 Dark channel prior · IEEE TPAMI

Canty & Nielsen 2008 IR-MAD · Remote Sensing of Environment

Oquab et al. 2024 DINOv2 (Meta AI) · TMLR

Haralick 1979 Texture features · Proceedings of the IEEE (not yet read)

Roberts et al. 2017 Blocked cross-validation · Ecography

BACKGROUND FOR THE PROPOSAL

Portalés-Julià et al. 2023 WorldFloods v2, cloud and water masks · Scientific Reports

Qu, Li & Huang 2026 AnyTime-CD · Remote Sensing of Environment

Shen et al. 2015 Missing-data reconstruction review · IEEE Geoscience and Remote Sensing Magazine

Bruzzone & Bovolo 2013 Change detection framework · Proceedings of the IEEE

METHOD ATTRIBUTIONS · BELOW THE JIF RULE

Breiman 2001 · Random forests · Machine Learning

McNemar 1947 · Paired proportions test · Psychometrika

Data: SpaceNet 8, Hänsch et al. 2022 (licence to confirm) · HRC_WHU (academic use only) · claims checked 2026-10-24

Evidence files

AGENTIC-GEOAI/CLAUDE/CLOUDRESEARCH/

`input_card.md`	the problem, scope, boundaries and decisions
`decision_log.md`	every accepted, changed and rejected item
`outputs/`	research questions, critiques, abstract, methodology v1 and v2
`evidence/`	measured results, literature summaries, research-question guide, reconstruction literature
`figures/ · presentation/`	the figures and this deck

CODE · GITHUB.COM/JROG378/GEOAI-ROADS (PRIVATE)

sn8_clouds.py (masks and the cloud rule) · sn8_refhaze.py (two-date de-hazing) · sn8_cloudref.py (HRC_WHU check) · the stage test notebooks